

A Penalty-Based Exterior-Interior Point Method for Stochastically Constrained Simulation Optimization

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Outline

- 1 *Introduction*
- 2 *Algorithm*
- 3 *Theoretical Results and Numerical Performance*
- 4 *Takeaway*

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What we did

Proposed a new algorithm for solving the following **stochastically constrained simulation optimization** problem of the form

$$\begin{aligned} \min_{x \in \Theta} \quad & f(x) := E[F(x, \xi_f)] \\ \text{s.t.} \quad & g^{[\ell]}(x) := E \left[G^{[\ell]} \left(x, \xi_g^{[\ell]} \right) \right] \leq 0, \quad \forall \ell = 1, \dots, q. \end{aligned}$$

Proved the almost sure convergence and expected convergence rate of the algorithm, and numerically validated its effectiveness.

Application example: NPC

Neyman–Pearson Classification (NPC)

$$\text{type I error} := E[I^+\{\tilde{h}(a; x)\} | b = -1],$$

$$\text{type II error} := E[I^+\{-\tilde{h}(a; x)\} | b = +1],$$

where $\tilde{h}(a; x)$ is a classifier parameterized by x , trained using a dataset a , $b \in \{-1, +1\}$ is a label, and $I^+\{\cdot\}$ is an indicator such that $I^+\{z\} = 1$ if $z \geq 0$ and $I^+\{z\} = 0$ if $z < 0$.

$$\min_x \quad \text{type II error}$$

$$\text{s.t.} \quad \text{type I error} \leq \iota,$$

where $\iota > 0$ is a given threshold.

What's new

$$\begin{aligned} \min_{x \in \Theta} \quad & f(x) := \mathbb{E}[F(x, \xi_f)] \\ \text{s.t.} \quad & g^{[\ell]}(x) := \mathbb{E} \left[G^{[\ell]} \left(x, \xi_g^{[\ell]} \right) \right] \leq 0, \quad \forall \ell = 1, \dots, q. \end{aligned}$$

- ▶ Sample average approximation (SAA) vs. Stochastic gradient descent (SGD)
- ▶ Unbiased stochastic gradient with bounded variance vs. Biased stochastic gradient with unbounded variance
- ▶ Explicit feasibility check vs. Implicit feasibility check

What's new

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New penalty functions

Define a family of parameterized **penalty functions** $\{r_\varepsilon(\cdot) : \varepsilon > 0\}$. We require them to meet some technical conditions.

Examples 2.1

- ▶ exponential penalty: $r_\varepsilon(y) = \exp(y/\varepsilon)$
- ▶ logarithmic (softplus) penalty: $r_\varepsilon(y) = \ln(1 + \exp(y/\varepsilon))$

Define a (deterministic) penalized problem:

$$\min_{x \in \Theta} h_\varepsilon(x) := f(x) + \sum_{\ell=1}^q r_\varepsilon \left(g^{[\ell]}(x) \right).$$

Denote by $x_\varepsilon^* := \arg \min_{x \in \Theta} h_\varepsilon(x)$ the penalized solution.

Convergence for the deterministic problem

Proposition 2.2

Given any positive sequence $\{\varepsilon_k\}$ such that $\varepsilon_k \rightarrow 0$. Applying the proposed penalty functions. Then there exists an integer N such that for all $k \geq N$, $\max_{\ell} g^{[\ell]}(x_{\varepsilon_k}^*) < 0$, and $\lim_{k \rightarrow \infty} f(x_{\varepsilon_k}^*) = f(x^*)$.

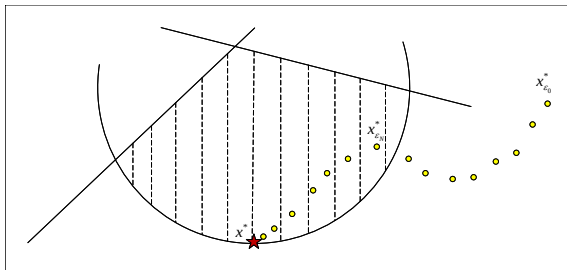


Figure 1: Illustration of the convergence behavior of the penalized problem

Multi-timescale stochastic approximation (SA)

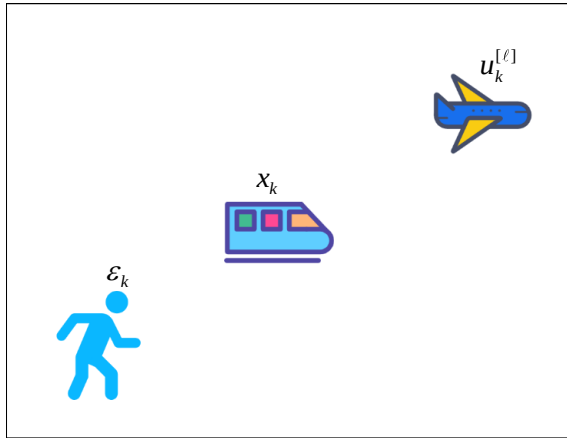


Figure 2: Multi-timescale SA algorithm design

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Convergence

Theorem 3.1

Suppose $h_\varepsilon(\cdot)$ is **strictly convex**. Under appropriate conditions (where the stochastic gradients are asymptotically unbiased and their variances can be increasing), the sequence $\{f(x_k)\}$ almost surely converges to $f(x^*)$; i.e.,

$$f(x_k) \rightarrow f(x^*) \text{ as } k \rightarrow \infty \text{ w.p.1.}$$

Rate of convergence

Applying logarithmic penalty functions; objective function $f(\cdot)$ is **strongly convex**, constraint $g^{[\ell]}(\cdot)$ is convex.

- ▶ Step sizes: $\alpha_k, \beta_k, \gamma_k$
- ▶ Decreasing rate of biases: b_k
- ▶ Increasing rate of variances: σ_k

Theorem 3.2

Under appropriate conditions,

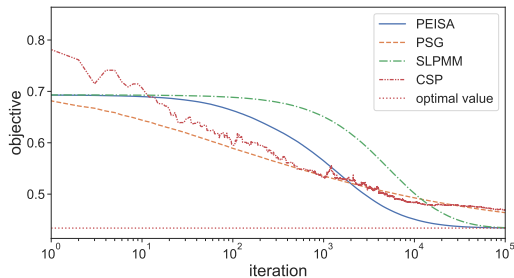
$$\begin{aligned} & \mathbb{E}[f(x_k) - f(x^*)] \\ &= O(b_k) + O\left(\frac{\beta_k \sigma_k^2}{\alpha_k \varepsilon_k^3}\right) + O\left(\sqrt{\frac{\alpha_k}{\varepsilon_k}} \sigma_k\right) + O\left(\frac{\gamma_k}{\beta_k \varepsilon_k}\right) + O\left(\varepsilon_k \ln\left(\frac{1}{\varepsilon_k}\right)\right). \end{aligned}$$

Rate of convergence (cont'd)

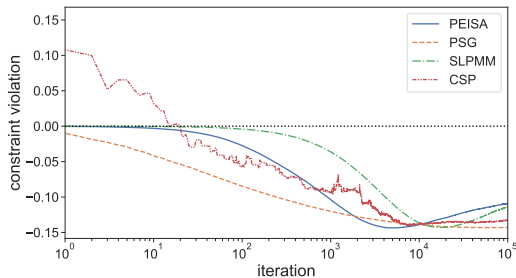
Use unbiased stochastic gradient estimators with bounded variances

- ▶ The optimal solution x^* is on the boundary of the feasible region:
Rate $O(k^{-1/12})$
- ▶ The optimal solution x^* is in the interior of the feasible region:
Rate $O(k^{-1/4})$
- ▶ ε -suboptimal solution: Rate $O(k^{-1/3})$

Numerical experiment



objective/iteration



constraint violation/iteration

Figure 3: Performance of comparison algorithms on the NPC problem, based on 30 independent runs

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Takeaway

- ▶ A Novel penalty-based exterior-interior point method
- ▶ Multi-timescale SA coordinates the multi-layer loop problem
- ▶ Explore more scenarios of constrained optimization

*Thank you for your time, and welcome your
comments and suggestions!*

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